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PROFESSIONAL SUMMARY

Computer Science student with hands-on production engineering experience in **system reliability, monitoring, and deployment**. Deployed and maintained live backend services with **Sentry error monitoring** and **Docker-based environment consistency**, resolving **API rate-limiting** and stability issues under real production traffic. Strong foundation in networking, operating systems, and distributed systems, backed by hands-on **cloud and CI/CD** experience (AWS, Docker) — aiming to grow into a Site Reliability Engineering role focused on building resilient, observable, and scalable infrastructure.

EDUCATION

Canara Engineering College, Mangalore

B.E in Computer Science & Engineering

CGPA: 8.26

2026

KSEAB (SSLC/PUC)

SSLC: 88.96% — PUC: 92.96%

EXPERIENCE

Orbit — Intern, Technical Team Member

Feb 2026 – Present

- Deployed and maintained a production backend (Node.js/Express) on Render with **Sentry error monitoring** and **Docker-based environment consistency**, supporting a live React Native/Expo mobile client.
- Diagnosed and resolved **API rate-limiting** issues surfaced in production, restoring stable service under real user load.
- Engineered real-time infrastructure using **WebSockets** for chat and geolocation-based event discovery, keeping connections stable under concurrent real-time traffic.
- Shipped mobile builds via **Expo Application Services (EAS)** and merged **5+ pull requests** into the production codebase, covering backend fixes and feature rollouts.

ACHIEVEMENTS & LEADERSHIP

Core Unleashed Hackathon — Team Lead, “Digital Sentinels”

Apr 2025

- Ranked among the **Top 10 teams** out of **72 teams** and pitched our solution to the final judging panel after advancing through multiple evaluation rounds.
- Led a team to build PII Shield, a privacy-first PII redaction tool, in a **36-hour hackathon**, driving the architectural decision to run **OCR (Tesseract) fully on-device** to eliminate cloud dependency and guarantee zero data retention.
- Engineered a **stateless backend** with a deterministic finding-ID system and a **3-tier bounding-box** fallback (exact character → line-level → sidebar-only) to keep redaction accurate despite imperfect OCR spatial mapping.

Feature Development & Deployment

Ongoing

- **Merged 5+ pull requests** into Orbit’s production codebase, shipping UI improvements, new features, and backend fixes.
- Redesigned the landing page and signup flow UI, and built a **location-picker dropdown that reverse-geocodes** a user-dropped map pin into an address.
- Implemented **event creation with feed-level visibility** and resolved **API rate-limiting** issues to keep the mobile client responsive under load.

PROJECTS

HVSapp — Healthcare Handoff Validation Platform

FastAPI, React Native, PostgreSQL, Celery

- Decoupled an **ASR transcription pipeline into async Celery task queues** to prevent blocking of real-time request handling, improving system throughput under load.
- Applied targeted **database indexing** and async task decoupling to improve API responsiveness under concurrent clinical workflows.
- Implemented PHI field encryption and medication safety tracking via Alembic migrations, structuring the schema around clear domain boundaries (auth, medication, patients, transcription).

GraphSentinel — Graph Neural Network Cyber Defense System

Python, PyTorch Geometric, GraphSAGE

- Designed and trained a GraphSAGE-based intrusion detection model on CICIDS-2017 and optimized inference to **~120ms per graph** to fit a production real-time detection budget of 5 seconds.
- Achieved **99.57% accuracy** and **0.9975 AUC-ROC** after resolving severe class imbalance (128K vs. 1,956 samples) using hybrid under-sampling with SMOTE and weighted cross-entropy loss.

TECHNICAL SKILLS

- **DevOps & Monitoring:** Docker, Git/GitHub, CI/CD basics, Sentry, Linux Administration, WebSockets, Celery
- **Backend & Cloud:** Node.js, Express, FastAPI, Flask, AWS (EC2, S3, VPC), PostgreSQL, MongoDB, Redis
- **Languages:** Python, Java, JavaScript, SQL
- **ML & Data:** PyTorch, PyTorch Geometric (GraphSAGE), Scikit-learn, OpenCV, MediaPipe
- **Frontend & Mobile:** React.js, React Native, Redux, Tailwind CSS